

Research paper

Understanding the barriers to recycling critical raw materials for the energy transition: The case of rare earth permanent magnets

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ABSTRACT

Rising geopolitical tensions and the effects of the Covid-19 pandemic have intensified concerns about securing access to the critical raw materials (CRMs) that are needed for the energy transition. A group of 17 elements referred to as rare earth elements (REEs) are among the CRMs for which security of supply is increasingly becoming a matter of priority for governments around the globe. REEs are particularly important in the production of rare earth (RE) permanent magnets, which due to their properties are key components of green energy technologies. While it is generally acknowledged that recycling can help mitigate supply shortage risks for REEs, a full recycling chain has not yet been established in the EU. Focusing on RE permanent magnets, this study provides a qualitative-exploratory analysis of barriers to establishing full supply chain recycling processes in the EU. Data have been collected through in-depth interviews with industry experts from all steps of the magnets' value chain as well as academics. According to our empirical findings, major barriers include limited information about the type of magnets included in end-of-life products, lack of recycling targets, lack of ecodesign requirements, difficulty in moving products across borders, lack of certification systems, high costs involved in the recycling processes, competition with magnets sourced from non-EU countries and missing segments of the RE value chain. Addressing the multitude of barriers in place would require coordinated action across several policy fields.

1. Introduction

The transition from a fossil fuels-based to a climate-neutral economy will entail an exponential increase in consumption of several critical raw materials (CRMs), including rare earth elements (REEs) (OECD, 2019; IEA, 2021; Gregoir et al., 2022). REEs comprise 17 elements of the periodic table (the 15 lanthanides, scandium and yttrium) that share some unique physical characteristics. Some of them are important for the manufacturing of neodymium-iron-boron (NdFeB) magnets that are extensively used in a wide range of applications, including low-carbon technologies such as electric vehicles (EVs) and wind turbines (Kalvig 2022; Gielen and Lyons, 2022; Balaram, 2019). Driven by the progressive market penetration of these low-carbon technologies, the global demand of rare earth (RE) permanent magnets – and hence of REEs – is set to surge over the coming years. For instance, Carrara et al. (2023) estimate that neodymium and dysprosium demand will increase by 4–5 times by 2030, and by 11–13 times by 2050. At the same time, the

current supply is highly concentrated, as China is responsible for 100 % and 85 % of the markets for processed heavy REEs and light REEs respectively (European Commission, 2023a). In addition, the country accounts for 85–90 % of overall NdFeB magnets production (Carrara et al., 2023), and exports in the EU 16 thousand tonnes of NdFeB market each year – that is, 98 % of the overall EU market (Gauß, et al., 2021). In the midst of rising geopolitical tensions and supply chain disruptions, these market conditions are urging governments around the globe – including in the EU – to re-assess their REEs sourcing options, and to adopt policies to secure their supply (IEA, 2023; European Commission, 2023b).

The magnitude of the global increase in the demand for REEs will inevitably require additional mining capacity in the future (Balaram, 2019; Habib and Wenzel, 2014). But while potential to expand primary production exists in the EU (Goodenough et al. 2016), structural inefficiencies of the mining sector, regulatory hurdles and public scepticism make the exploitation of such potential very difficult in the

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short-to-medium term (Righetti and Rizos, 2023; Stegen, 2015). In this context, boosting REEs' circularity and recycling – particularly for permanent magnets – can therefore play an important role in decreasing the pressure on additional primary production (Månberger, 2023). Recycling REEs also offers economic and environmental benefits compared to primary resource extraction (Fujita et al., 2022; Dutta et al., 2016). Indeed, rare earths mining is traditionally associated with significant environmental and social impacts (Balaram, 2019; Golroudbary et al., 2022; Zapp et al., 2022). With regard to magnets more specifically, recycling was demonstrated to have substantially lower environmental impacts than virgin magnets production (Jin et al., 2018). In general, recycling can also help address security of supply concerns that have intensified following the Covid-19 pandemic and the geopolitical tensions due to the war in Ukraine (Cheramin et al., 2021; Righetti and Rizos, 2023).

When it comes to rare earth magnets, two major recycling pathways exist. The first involves 'indirect' recycling via traditional hydrometallurgical and pyrometallurgical recovery processes, while the second concerns 'direct' (also known as 'magnet-to-magnet') recycling via hydrogen-based processes. The former route allows for a high degree of flexibility, as it enables to retrieve individual REEs which can then be used for a wider range of applications than permanent magnets only (Binnemans et al., 2021). However, pyro- and hydrometallurgical processes make intensive use of energy, chemicals and water (Binnemans et al., 2013; Firdaus et al., 2016; Takeda and Okabe, 2014). The second pathway involves the hydrogen decrepitation of end-of-life (EoL) magnets into a powder subsequently used to manufacture new magnets (a process often referred to as 'hydrogen processing of magnets scrap', or HPMS) (Yang et al., 2017; Walton et al., 2015; Zakotnik et al., 2004). While coming with its own set of technical constraints – explored at greater length in Section 4.4 – this process allows for a shorter and less energy and chemical-intensive recycling of magnets (Yang et al., 2017; Habibzadeh et al., 2023), and has been making progresses towards commercialisation (Burkhardt et al., 2023).

Existing research indicates that there is considerable potential for magnet recycling to cover a share of future demand for REEs (Gregoir et al., 2022; Reimer et al., 2018; Rizos et al., 2022). For instance, Schulze and Buchert (2016) calculate that between 18 % and 22 % of global neodymium and praseodymium and 20–23 % of dysprosium and terbium demand from NdFeB magnets production could be met by secondary supply by 2030. At the EU level, Ciacchi et al. (2019) estimate that up to 50 % of annual EU consumption of neodymium could be covered by secondary sources, predominantly from RE magnets.

Still, several bottlenecks hinder the development of a full recycling value chain in the EU (Ilankoon et al., 2022; Fujita et al., 2022; Golev et al., 2014). While the technical difficulties of recycling are covered extensively in the literature (e.g. Jowitt et al., 2018; Yang et al., 2017; Binnemans et al., 2013), only a few studies have assessed the full spectrum of existing barriers. The present study contributes to the literature by providing an in-depth analysis of these barriers. Though different types of RE magnets' chemistries exist (including samarium-cobalt (SmCo) based ones), this research focuses on NdFeB magnets (hereinafter simply referred to as RE magnets) as the most widely employed type of magnet in the market, notably for low-carbon technologies (Kalvig 2022; Ganguli and Cook, 2018; Bailey et al., 2017).

Our research question is: *What are the key barriers to developing a recycling value chain for RE permanent magnets in the EU?* To the best of our knowledge, this is the first qualitative- exploratory study on barriers to developing an EU recycling value chain for RE permanent magnets using data from in-depth interviews with experts from the full RE permanent magnet value chain. Previous studies focused on other regions (i.e. Brazil, see Raspini et al., 2022) or used an industry survey with ranking questions (Ilankoon et al., 2022).

This paper starts with a review of the literature on RE magnets' recycling to identify barriers to recycling that have been previously documented (Section 2). It then describes the research methodology

(Section 3) followed by a presentation of the empirical results (Section 4). The key findings are then discussed in Section 5 and finally Section 6 presents the conclusions and suggestions for future research.

2. Literature review

This section identifies barriers to the recycling of RE magnets that have been raised in the literature. Inspired by the work of various authors, (see, for example, Rizos and Bryhn, 2022; Raspini et al., 2022; Ilankoon et al., 2022), we identify the following major categories of barriers: lack of sufficient policies and regulations, challenges of an economic and financial nature, restrictions stemming from supply chains, and technological obstacles (see Table 1).

From a *policy* perspective, the lack of sufficient public support in the form of financial incentives to boost recycling operations and the development of secondary markets is a first barrier documented in the literature (Tsamis and Coyne, 2015; Binnemans et al., 2021; Ilankoon et al., 2022). Another barrier raised is that current support through R&D programmes is insufficient to scale up innovative recycling practices (Tsamis and Coyne, 2015). For instance, it has been highlighted that further R&D support is needed to promote a common approach to quantifying the environmental benefits of magnet recycling (Frost et al., 2021), to develop new RE separation and purification technologies (Ilankoon et al., 2022) and to test these technologies in pilot plants (Schüler et al., 2011).

Another group of missing policies and regulations relates to the applications of RE magnets. Scholars emphasise that the design of applications with magnets does not facilitate their easy recycling at the EoL stage (Schüler et al., 2011; Tsamis and Coyne, 2015; Burkhardt et al., 2020). Recycling operations are often hindered by the lack of labels on products that inform on the type of magnets used, their grade, their chemical compositions and the types of coatings they use (Burkhardt et al., 2020; 2021). The lack of sufficient policies to support effective collection of EoL applications containing RE magnets can lead to uncontrolled disposal and reduce the availability of material for recycling (Schüler et al., 2011; Haque et al., 2014; Gregoir et al., 2022).

Challenges of an *economic and financial* nature preventing the development of a circular economy for RE magnets were also identified in the literature. First, due to the high price volatility of REEs and RE magnets, future revenues from recycling are unpredictable (Golev et al., 2014; Tsamis and Coyne, 2015; Müller et al., 2016). Companies are also reluctant to invest in recycling processes due to uncertainties about their financial profitability (Raspini et al., 2022; Binnemans et al., 2021). One notable constraining factor is the difficulty for recycled magnets to compete with cheaper imported magnets using virgin material (Diehl et al., 2018; Geng et al., 2021; Nguyen et al., 2017). Moreover, disassembling the EoL products and extracting the magnets can be a costly process involving significant manual work (Sprecher et al., 2014; Burkhardt et al., 2020; Fujita et al., 2022).

Barriers also emerge from the RE magnet *supply chain*. One frequently stated barrier in this category relates to the current low volumes of RE magnets from EoL products reaching recyclers (Nlebedim and King, 2018; Gregoir et al., 2022; Haque et al., 2014). Forecasting future demand and recovery rates of RE magnets' applications in Europe, Reimer et al. (2018) point out that developing a circular economy for RE magnets can be constrained by low quantities of magnets' scrap available for the recycling plants. Nevertheless, it is likely that RE recycling will become more feasible with larger waste streams in the future (van Nielen et al., 2023).

There is also high uncertainty regarding future waste streams of RE magnet applications, particularly concerning their volume, type and accessibility (Schüler et al., 2011; Sprecher et al., 2014; van Nielen et al., 2023). As stated above, recyclers frequently do not have key information about magnet content and location. As the example of the automobile sector shows, this lack of information from manufacturers often prevents recyclers from having access to a sufficient number of magnets (Jönsson

et al., 2020). In addition, a further barrier to increased circularity along the supply chain for RE magnets is its underdeveloped collection and sorting infrastructure. Insufficient collection and sorting facilities, paired with restricted access to recycling loops of other metals or products containing REEs may lead to RE magnet feedstock ending up in landfills and incineration (Li et al., 2022; Fujita et al., 2022; van Nielen et al., 2023).

From a *technology* perspective, several barriers arise from the characteristics of RE magnet applications and the dismantling process. Due to the low concentration of RE magnets in several products, such as hard disk drives or other consumer electronics (Jowitt et al., 2018), they often have to be extracted manually, which is highly labour intensive if material losses are to be avoided (Diehl et al., 2018; Sprecher et al., 2014). The fact that glues or plastics are often applied to RE magnets further complicates their extraction (Li et al., 2019). It has also been highlighted that most scrap separation technologies for post-consumer applications are designed to only detach ferrous scrap from non-ferrous scrap, and are therefore not suitable for extracting RE magnetic material (Yang, et al., 2017).

In addition to the challenges related to magnets' extraction and separation, technical barriers reflecting the limitations of existing recycling technologies have also been raised in the literature. For example, the hydrometallurgical recycling approach generates large amounts of hazardous chemicals and wastewater, and requires various processing steps to obtain secondary magnets (Binnemans et al., 2013; Golev et al., 2014; Sprecher et al., 2014). On the other hand, the pyrometallurgical approach requires less processing steps but is highly energy intensive and generates large amounts of solid waste (Sprecher et al., 2014; Yang et al., 2017; Habib, 2019). While the HPMS process is generally less polluting, it also faces some technical difficulties. Notably, the process requires EoL magnets to be of relatively high quality (i.e., neither contaminated nor oxidised) to operate efficiently and prevent significant performance losses, and can therefore require several purification steps (Binnemans et al., 2013; Yang et al., 2017; Jönsson et al., 2020). Implementing this process at scale would require improving current collection and sorting processes as also discussed above (Burkhardt et al., 2023).

3. Methodology

A qualitative-exploratory research method was adopted for this study. This method can be justified on several grounds. First, exploratory studies are well-suited for addressing 'what' questions that seek to develop new knowledge and identify areas requiring further research (Yin, 2018). Second, such studies can provide new insights on topics that are under researched and for which empirical evidence is lacking (Swedberg, 2020). Third, the qualitative research approach is useful for developing 'an in-depth understanding of real-world problems' and to 'explore or explain phenomena in real-world contexts' (Korstjens and Moser, 2017, p. 275). Finally, the qualitative-exploratory research approach allows the use of less standardised questions to facilitate the collection of 'rich and explanatory in nature' qualitative data (Mack et al., 2005, p. 4). Interviews were used as the main format for collecting in-depth perspectives and data on the real-world challenges for developing recycling processes for RE permanent magnets in Europe (Mack et al., 2005; Menon, 2019).

The development of the sample was based on a combination of the purposive sampling (Saunders et al., 2009) and snowball sampling techniques (Lune and Berg, 2017). This involved leveraging the network of researchers including the partners in the INSPIRES (2023) project¹ and asking the interviewees to provide referrals for experts in the field

Table 1

Overview of the different barriers in the literature.

Category	Barrier	Ref.
Policy	Limited financial incentives for secondary markets	Tsamis and Coyne (2015); Binnemans et al. (2021); Ilankoon et al. (2022)
	Lack of R&D funding	Tsamis and Coyne (2015); Frost et al. (2021); Ilankoon et al. (2022); Schüler et al. (2011)
	Lack of ecodesign requirements	Schüler et al. (2011); Tsamis and Coyne (2015); Burkhardt et al. (2020)
	Absence of legal requirements for labels or markings	Burkhardt et al. (2020); Burkhardt et al. (2021)
	Insufficient policies on collection of EoL products	Schüler et al. (2011); Haque et al. (2014); Gregoir et al. (2022)
Economic	Volatile rare earth prices	Golev et al. (2014); Tsamis and Coyne (2015); Müller et al. (2016)
	Uncertainty about the profitability of magnet recycling	Raspi et al. (2022); Binnemans et al. (2021)
	Competition with primary magnets	Diehl et al. (2018); Geng et al. (2021); Nguyen et al. (2017)
	Costly extraction of EoL magnets	Sprecher et al. (2014); Burkhardt et al. (2020); Fujita et al. (2022)
Supply chain	Lack of stable supply of EoL products	Nlebedim and King (2018); Gregoir et al. (2022); Haque et al. (2014); Reimer et al. (2018); van Nielen et al. (2023)
	Unpredictability about future volumes of EoL permanent magnets	Schüler et al. (2011); Sprecher et al. (2014); van Nielen et al. (2023)
	Uncertainties about permanent magnet use in products	Jönsson et al. (2020)
	Underdeveloped collection and sorting infrastructure	Li et al. (2022); Fujita et al. (2022); van Nielen et al. (2023)
Technology	Difficult extraction of EoL magnets	Jowitt et al. (2018); Diehl et al. (2018); Sprecher et al. (2014); Li et al. (2019); Yang et al. (2017)
	Environmental impacts and long processing steps of hydrometallurgical recycling processes	Binnemans et al. (2013); Golev et al. (2014); Sprecher et al. (2014)
	Environmental impacts of pyrometallurgical recycling processes	Sprecher et al., (2014); Yang et al., (2017); Habib, (2019)
	Limitations of the HPMS process	Binnemans et al. (2013); Yang et al. (2017); Jönsson et al. (2020); Burkhardt et al., (2023)

(Rizos and Bryhn, 2022). To address the research question, the research team sought to include experts with extensive knowledge on the subject from both industry and academia (Roberts et al., 2023). To be more specific, the selection of the sample was based on the following criteria: i) including representatives from companies with direct engagement in RE recycling processes either at the R&D stage or at the stage of upscaling existing solutions in the market (Roberts et al., 2023), ii) having at least one company from each step of the RE magnets' value chain (mining and refining of rare earths, manufacturing of permanent magnets, collection of EoL products, dismantling, recycling, production of new magnets, use in new products) (Serna-Guerrero et al., 2022), iii) including academics with at least one research publication in the field of circularity of REEs. As shown in Table 2, 18 organisations from industry and academia were included in the sample, while 21 experts in total participated the interviews since in three cases the organisations had two representatives. Fig. 1 presents an overview of the RE permanent magnets value chain and the number of sampled organisations in each

¹ INSPIRES is a project co-funded by the EU that aims to develop recycling processes for rare earth permanent magnets, see: <https://www.inspires-magnet.eu/>.

Table 2
Interviewed experts.

Sample size (n = 18)		
Interview number	Sector	Position
1	Industry: Manufacturing of Permanent Magnets	Engineer
2	Academia	1. Professor 2. Assistant Professor
3	Industry: Recycling & Refining	Director
4	Industry: Recycling	Head of Department
5	Academia	Scientific Director
6	Academia	Professor
7	Industry: Data Analysis & Certification	Director
8	Industry: Dismantling & Extraction of Permanent Magnets	Analyst
9	Academia	Researcher
10	Industry: Manufacturing of Motors	Head of R&D
11	Industry: Waste Collection	Manager
12	Academia	1. Researcher 2. Researcher
13	Industry: Manufacturing of Motors and Automotive systems	Senior Expert
14	Industry: Manufacturing of Home Appliances	1. Head of R&D 2. Project Manager
15	Academia	Assistant Professor
16	Academia	Researcher
17	Industry: Mining, Refining & Recycling	Vice President
18	Industry: Mining	Board Member

step.

Interviews were conducted from October 2021 to February 2023 during the implementation of the INSPIRES project. In order to participate in the assessment, the experts received via email an interview package which featured the questionnaire and a form describing the purposes of the study and explaining that information provided would be anonymised in the paper. As discussed above, the interviews and the questionnaire followed an open format to enable the collection of elaborated responses and rich qualitative data. The questionnaire started with a general question inviting the experts to describe their experience with recycling and circularity processes for REEs and permanent magnets. It then asked the experts to describe in detail what are the key barriers (policy/financial/technological/market) to developing a recycling value chain in the EU. Having agreed to an interview, a link for online participation was sent to the experts. The interviews lasted between 45 and 90 min.

The interviewees all agreed to having the online interviews recorded. The interview recordings were stored in SharePoint to which only the researchers had access to. Verbatim transcriptions were prepared for all interviews to ensure that detailed accounts of all points raised would be taken into account in the analysis (Poland, 1995). This process resulted in 122 pages of transcript. Data were grouped first under the main categories of barriers described (see Section 2) and second under a number of codes (n=29, see Table 3) to facilitate the analysis of extensive textual qualitative data (Lune and Berg, 2017). The first group of codes used was retrieved from the literature review presented in Section 2. For barriers that have not yet been documented in the

literature the authors used the descriptive coding technique to identify the relevant codes which involved summarising “in a word or short phrase...the basic topic of a passage of qualitative data” (Saldaña, 2013, p.88). The coded data were then transferred to an Excel document and filtered under the categories of barriers and the different codes to facilitate the analysis. This facilitated the identification of common barriers across the interviews. Fig. 2 presents an overview of the different steps carried out for the analysis.

4. Results

4.1. Policy barriers

One of the most pressing policy-related barriers mentioned during 9 interviews is the absence in the EU legislative framework of specific rules requiring the provision of information about the magnets included in different EoL applications. Due to the lack of any label or marking when EoL applications are collected the magnet scrap has to be extracted from the application, sorted and analysed to verify the type of magnet and material composition. This uncertainty makes economic efficiency and scaling up of recycling processes more burdensome. As noted by 1 of our interviewees (Manager, Industry), “*having clear markings about rare earth content would strongly incentivise companies to recover permanent magnets and close their loop*”. Interviewees recommended that digital technologies and the introduction of a mandatory product passport, as it will be required at the EU level for batteries, could provide a solution for this challenge and support the collection of data about rare earth magnets along supply chains.

A reoccurring barrier mentioned in 7 interviews referred to the limited financial incentives currently available in Europe to support secondary markets for rare earth magnets. While there have been several developments in the technological arena of RE recycling, there is still an important competitive disadvantage in secondary production compared to primary production. The need to increase the financial support provided to RE recycling projects through public R&D programmes was remarked on during 6 interviews. As noted by the experts, such funds should have the two-fold objective of bringing existing recycling technologies into the market while at the same time channelling financial support for the breakthrough innovations that are still required for the sector. It was emphasised that there is considerable bureaucracy involved in applying for EU and national R&D funding opportunities, which is often discouraging for companies and especially for small and medium-sized enterprises.

In 5 interviews experts discussed barriers stemming from the lack of specific targets supporting the recycling of REEs. Specifically, 3 interviewees proposed that the introduction of recycled content targets for new RE products would provide a boost to the industry. Inspired by the mandatory recovery targets introduced for materials (e.g. cobalt, lithium and nickel) in EoL batteries through the EU Batteries Regulation, 2 other experts indicated that there is also a need for RE-specific recovery targets to drive the development of an EU recycling industry. In both cases the experts noted that any such targets should be introduced through a step-wise process allowing the industry to prepare and develop the required facilities.

The importance of product design for downstream recovery of rare earth permanent magnets was noted in 4 interviews. Using the example of electric vehicle motors, 1 interviewee (Researcher, Academia)



Fig. 1. Overview of the RE permanent magnets value chain. This is simplified version of the chain that does not feature some steps upstream (e.g. separation of RE oxides). The numbers in brackets indicate the number of sampled organisations in each step of the chain. Notably, out of the 18 sampled organisations from industry and academia, 7 provided horizontal perspectives and are not counted in these numbers.

Table 3
List of barriers raised during the interviews.

Category	Barrier	Number of interviews
Policy	Absence of legal requirements for labels or markings	9
	Limited financial incentives for secondary markets	7
	Lack of R&D funding	6
	Lack of targets supporting recycling	5
	Lack of ecodesign requirements	4
	Lack of multilateral solutions	4
	Complex legal framework for intra-EU movement of EoL products	3
	Lack of certification systems	3
	Lack of a comprehensive EU industrial policy framework	3
	Insufficient policies on collection of EoL products	3
	Lack of extended producer responsibility requirements	2
Economic	Costly extraction of EoL magnets	9
	Competition with primary magnets	8
	Competition with non-EU suppliers	8
	Low rare earth prices	3
	Uncertainty about the profitability of magnet recycling	3
	Volatile rare earth prices	2
	Missing market for recycled magnets	2
	High energy prices	2
Supply chain	Missing segments in the value chain	11
	Uncertainties about permanent magnet use in products	8
	Reluctance to use permanent magnets with recycled content	6
	Underdeveloped recycling networks	6
	Lack of stable supply of EoL products	5
	Unpredictability about future volumes of EoL permanent magnets	5
Technology	Difficult extraction of EoL magnets	10
	Limitations of the HPMS process	8
	Lack of technical know-how in the EU	6
	Environmental impacts and long processing steps of hydrometallurgical recycling processes	2

mentioned that “an efficient way to extract the magnets out of the motors is lacking because the current design of these motors is not made for disassembly and recycling”. Hence, drawing lessons from the impact of ecodesign requirements introduced for electrical and electronic products through the EU ecodesign legislative framework, specific dismantling and recyclability requirements could be introduced for products with rare earth magnets according to the experts. The international dimension of RE recycling was also highlighted in the interviews. Four experts proposed that more work should be done to identify multilateral solutions given that supply chains for applications containing rare earth magnets are interconnected and global, and thus it is counterproductive to only seek domestic solutions. In this context, in parallel to developing its own recycling industry, the EU should seek bilateral co-operation to co-develop with other partners strategic projects along the supply chain and where possible multilateral solutions. Two specific areas mentioned where there is scope for more international cooperation are: improving transparency across the magnet supply chains and establishing a common methodology for measuring environmental footprint across supply chains. The difficulty of developing recycling processes for REEs is further exacerbated by the existing rules for intra-EU movement of EoL products and parts containing RE magnets according to 3 interviewees. The experts noted that the process of shipping such products that are considered as waste is complex and needs to be facilitated when high environmental standards can be showcased. It was also argued that the interpretations of waste policies across the EU can be inconsistent, and there are varied waste processing and collection approaches which create further obstacles to building RE recycling processes in Europe.

Another barrier mentioned by 3 interviewees concerns the lack of credible certification systems to help build confidence along the supply chain for recycled permanent magnets. One of the experts (Head of Department, Industry) stressed that there are “problems with the usage of recycled magnets in certain industries because they lack a certification system to assure the magnetic properties of the recycled materials”. Three experts identified the lack of a comprehensive EU industrial policy framework promoting all parts of the rare earth value chain including mining, refining and recycling as a barrier. According to 1 of them (Head of R&D, Industry), “in this EU policy domain there is a conflict between sustaining high environmental standards while at the same time significantly increasing own supply of resources”. Policies on the collection of EoL products

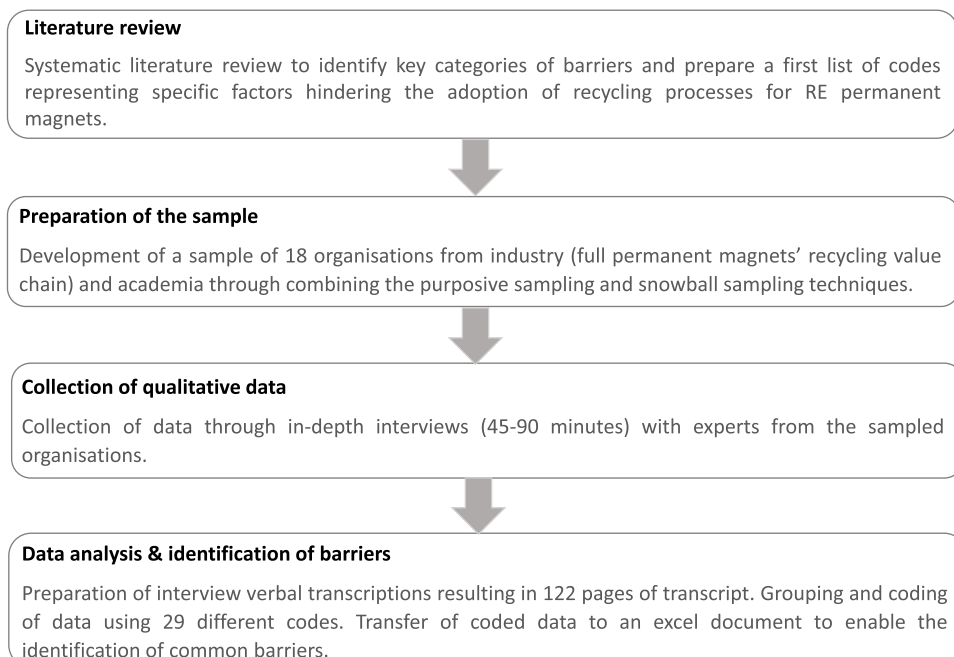


Fig. 2. Overview of the research steps applied for this study.

containing RE permanent magnets are insufficient according to 3 experts. As noted by the interviewees, there is only limited collection of small electronic appliances containing permanent magnets partly due to the low societal awareness of the impacts of inappropriate management of these devices. Factors such as proximity to collection centres and familiarity with the recycling process may influence consumers' decision to recycle their devices. Concerns about data security can also be a factor affecting the mismanagement of electronic devices containing magnets as shown by the example of hard disk drives which are sometimes shredded for data security reasons. Finally, during two interviews it was highlighted that the current EU policy framework lacks specific extended responsibility requirements for permanent magnets similar to the ones that are in place for batteries.

4.2. Economic barriers

With regard to the economic nature barriers, the major issue – raised by 9 consulted experts – refers to the high costs involved in extracting magnets from EoL applications. As noted by these experts, retrieving EoL magnets requires substantial work from recyclers, who are indeed forced to disassemble or dismantle the products, often manually, to access the magnets. This is partly due to the relatively low concentration of magnetic material in products; owing to their magnetic properties they are firmly embedded in the ferrous fraction of products. In addition, the use of glues often further complicates their extraction. In electric vehicles, for example, magnets are typically glued in the rotors of the electro-motor, so their extraction requires intensive manual work or having in place automated processes involving multiple steps. Added to this, the lack of labelling requirements mentioned above often prevents recyclers from knowing beforehand where the magnets are located. Since the design of products can vary considerably even within the same product group, recyclers are therefore forced to break products apart to locate and access magnets. As a result, magnet extraction is a highly time and labour-intensive activity, which adds considerable costs and affects the economic viability of the whole recycling process. In this respect, providing financial incentives for recycling, or enforcing stricter and homogeneous 'design for recycling' requirements were raised as possible countermeasures to alleviate the problem.

The high costs involved in the recycling processes of magnets – and particularly in the extraction step – raises a second key economic barrier, namely the difficult competition with the production of primary magnets. In fact, as noted by a representative from the recycling sector (Head of Department, Industry), *"recycled magnets are likely to be more expensive than those produced via primary materials, and most companies are still not willing to pay more for them"*. This concern – shared by 8 experts – applies in particular to the early development stages of recycling, before economies of scale can improve the profitability of the whole process. In addition, 8 other respondents also highlighted that competition is particularly harsh versus magnets sourced from non-EU countries and particularly China which has long established an industrial strategy to support the industry. In turn, such challenging market competition makes it difficult to develop a business case around and attract investments into magnet recycling in the EU.

A further barrier refers to the price dynamics of rare earths, which according to the experts is not conducive to the creation of business cases around magnets recycling. More specifically, 3 interviewees underlined that – at the time of the interviews – the prices of rare earths were too low to incentivise their recycling, especially given the relatively low amounts contained in some products such as in small electronic appliances or internal combustion engine vehicles. As pointed out by 1 interviewee, for magnet recycling to be profitable the price of the materials should be high enough to justify their extraction from EoL products, which as stated above is per se a highly costly activity. Added to this, 2 experts noted that the high price volatility of these materials also represents a constraining factor, as it creates uncertainty over future revenue stream and limits planning opportunities. This price factor,

coupled with high extraction costs, in turn creates uncertainty about the profitability of the whole recycling process – an issue raised by 3 experts.

Other economic barriers raised during the interviews include, on the demand side, the absence of an established EU market for recycled magnets. Indeed, as stated by two experts the lack of EU demand for (more expensive) domestically produced secondary materials further increases uncertainty on the business case for RE magnets recycling. In addition, the high energy prices following the energy crisis were also indicated by 2 interviewees as negatively affecting the economics of RE magnets' recycling processes.

4.3. Supply chain barriers

Interviewees noted supply chain barriers on various fronts. The fact that the EU currently lacks or has very limited capacities for various segments of the RE magnets value chain was remarked on during 11 interviews. In addition to the lack of domestic mining capacities, the EU is missing the RE refining segment of the chain for which China holds the largest capacities worldwide. According to one expert, difficult market conditions and relatively high energy prices are the major factors behind the lack of RE refining in the EU. Along with refining, the European magnet recycling and magnet manufacturing industries are also very small and largely concentrated in China. As a result, the EU is lacking the know-how and often the skills to develop at scale the different value chain segments, including recycling. As noted by the interviewees, building a viable recycling industry for RE magnets requires having domestic capacities in place for all value chain segments if investment certainty is to be assured.

According to 8 experts, while RE magnets can be used in a variety of products present in the market from consumer electronics to white goods and e-bikes, there is uncertainty about their use in these products. This has an impact on the economic feasibility of the recycling process. To be more specific, due to a combination of factors including market trends, technology developments, and cost reasons the use of RE magnets can vary significantly even within the same product group. Given the absence of any marking or information requirement specific for magnets, identifying whether these products contain RE permanent magnets requires a manual and labour-intensive dismantling process. The reluctance of magnet producers to use recycled content was perceived as a barrier by 6 experts. As noted during the interviews, companies prioritise the use of virgin material inputs due to concerns about the performance of magnets with recycled content. The absence of a credible certification scheme to provide a form of assurance about the qualities of recycled magnets contributes to this barrier.

A further supply chain barrier discussed during 6 interviews pertains to the underdeveloped recycling networks for RE permanent magnets. The recycling landscape for magnets is still fragmented with no complete recycling loops involving all actors, namely original equipment manufacturers, dismantlers and recyclers of different components. As noted by 1 expert, (Head of R&D, Industry), *"a situation in which the whole supply chain for recycled permanent magnets cooperates and shares knowledge currently only exists at small scale and during pilot projects"*. The low numbers of EVs and wind turbines with high magnet content reaching the EoL stage is among the key factors that contribute to this challenge. The existence of various different collection schemes and infrastructures across the EU for EoL products containing magnets further complicates the development of complete recycling networks. A closely linked barrier highlighted during 5 interviews concerns the lack of a stable supply of EoL products containing permanent magnets. As stressed by the experts, the volumes currently available for recycling are too small to incentivise the scaling-up of recycling processes. One interviewee (Assistant Professor, Academia) noted that *"most appliances currently available for recycling are small ones that contain relatively small amounts of rare earth magnets"*. Added to this, extracting the magnets from the several different products containing permanent magnets can often be technically challenging and labour intensive.

Five experts identified the unpredictability about future volumes of EoL permanent magnets as a supply chain-related challenge. Although future flows of EoL EVs and wind turbines are expected to significantly increase in the coming years and decades, it is difficult to make precise estimates about future demand and supply. As innovation trends constantly evolve, there are uncertainties with regard to the future pool of technology applications containing permanent magnets, their expected lifetime and the types and size of magnets used. All these create uncertainties about the future market for recycled magnets.

4.4. Technical barriers

Technical barriers also prevent the establishment of rare earths magnet recycling in the EU. The difficulty of extracting magnets from disposed applications was raised by 10 interviewed experts as a major barrier of technical nature. As explained above, magnets' physical properties and low concentrations in products, as well as the use of glues or the wide heterogeneity of product designs make magnet extraction a complex task for recyclers, particularly for smaller applications. This complexity often leads to magnets being damaged during the extraction process, thus to material losses. Developing advanced sorting and separation technologies was indicated as a possible solution to facilitate the extraction process, though 1 expert highlighted that these could raise further limitations, insofar as sorting and separation processes would need to be tailored to specific product groups, which have very diverse requirements. Adopting product designs conducive to magnet recycling was indicated by some as another way to improve the efficiency of the extraction step.

While extracting magnets tends to be particularly complex for smaller products, accessing larger magnets was also indicated as having its own challenges. For instance, 1 interviewee noted that retrieving large magnets from EoL wind turbines is complicated as these tend to be dangerous to handle when still magnetised; therefore, in these cases the extraction requires on-site demagnetisation or on-site pulverization (that is, the first step of the actual recycling process).

A range of different limitations of one specific recycling method – the HPMS – were mentioned by 8 interviewees as technical constraints. Notably, 4 of them underlined that the process requires high purity of feedstock to be efficient, and that different types of contaminants can affect the quality of the output. One of the major sources of contaminants indicated are magnet coatings, which are often used to protect magnets from corrosion. In some cases, the coatings can be easily removed in the pre-processing stage (via e.g. sand blasting). However, in other cases removing the coatings is challenging, since they decompose and remain as impurities throughout the process, with a negative impact on the quality of recycled magnets. In addition, the wide range of coatings employed requires recyclers to adopt different technologies to handle them, adding further complexity. Experts suggested that providing more information to recyclers – e.g. in the form of a label – regarding the type of coating used and preventing manufacturers from using coatings not compatible with the HPMS recycling process could help tackle this issue. Besides coatings, the glues used to anchor magnets in products were also mentioned as a possible source of contamination, as once burned they generate carbon that affects the quality of recycled material.

A second limit of the HPMS process, mentioned by 2 experts, is the difficulty of dealing with magnet oxidation. These experts noted that magnets tend to oxidise throughout their lifetime, and highly oxidised magnets can only be recycled by adding virgin materials to reach the required magnet density. This, however, leads to a downgrade in the magnetic properties of magnets, which can be particularly problematic for the recycling of low-grade magnets. Finally, 2 other experts pointed out that since the chemical composition of magnets tends to be highly product specific, the HPMS process can only be applied to magnets with a fairly homogeneous composition in order not to incur quality downgrades. For the same reason, magnets recycled through HPMS can only

be used for the same type of applications from which they were retrieved.

Another important barrier of technical nature, raised by 6 experts, refers to a general lack of technical know-how in the EU, particularly compared to China which has a long-established magnet production and recycling sector. Among the interviewees, 1 expert noted that the EU lacks the necessary engineering skills and expertise to develop a magnet recycling industry at scale, while 2 others noted the lack of familiarity within companies – and particularly among recyclers – on magnet categories and compositions as an issue.

Although the majority of recyclers included in the sample had experience (and elaborated) on the HPMS process, shortcomings of the hydrometallurgical recycling process of RE magnets were also raised during two interviews. In particular, one interviewee highlighted that this recycling technology involves several processing steps and therefore has costly infrastructural requirements, particularly compared with the HPMS process. Further, in line with findings from the literature, one other expert pointed out that hydrometallurgical recycling may have significant environmental impacts in the form of CO₂ and dust emissions.

5. Discussion

This study increases understanding of the intersecting factors of policy, economics, supply chains and technology that hinder the establishment of recycling processes for RE magnets in the EU. Interview findings validate several policy-related barriers documented in the literature, namely lack of public support in the form of R&D funding and financial incentives (e.g. Frost et al., 2021; Ilankoon et al., 2022), lack of ecodesign requirements (e.g. Tsamis and Coyne, 2015) and insufficient policies on collection of EoL products with RE magnets (e.g. Haque et al., 2014; Gregoir et al., 2022). A prime barrier, previously raised by Burkhardt et al. (2020); (2021) relates to the lack of legal rules on providing information in the form of labels about the magnets included in EoL applications. While corroborating this barrier, our findings go further in highlighting its importance for the economic feasibility of recycling processes. Notably, the recently adopted EU regulation on CRMs, the so-called European Critical Raw Materials Act, sets mandatory labelling requirements for a range of different applications including wind energy generators, electric motors, heat pumps, industrial robots, washing machines and vacuum cleaners (European Union, 2024). Still, the average lifetime of these applications often extends well beyond 10 years (Rizos et al., 2022) and thus there will be an extended period of time before they reach the EoL stage and the above labelling requirements benefit the recycling of RE magnets.

However, our empirical insights provide a number of policy-relevant novel contributions. One important barrier identified, not documented in the academic literature, refers to the lack of EU targets aimed at boosting both the recovery of REEs from EoL applications and the manufacturing of magnets with recycled content. This barrier, coupled with the absence of labelling requirements discussed above implies there is a need for an EU regulatory framework for RE magnets that on the one hand provides incentives for the development of recycling markets and on the other hand facilitates recycling through transparency obligations. For batteries the EU has put in place such a regulatory framework with concrete material recovery targets, minimum recycled content shares and information requirements (Rizos and Urban, 2024; Albertsen et al., 2021). A similar framework has been put forward for permanent magnets through the new European Critical Raw Materials Act (European Union, 2024) that anticipates the introduction of secondary legislation setting recycled content obligations and lays down labelling requirements as noted above. Material recovery targets for REEs could also be introduced in the future when the recycling market has matured through product-specific legislation. Our empirical findings indicate that these policies hold significant potential to support the development of a recycling chain in Europe, though given the time required for their

formal introduction and implementation their impact is not expected to be seen in the short-term.

Other key barriers stemming from the policy environment that are not observed in the literature are lack of credible certification schemes and complex legal rules for shipping EoL products with magnets within the EU for recycling. Given the fact that supply chains for rare earth magnets are global, expert findings also point to the need for more international cooperation and multilateral solutions for achieving circularity of magnets. Taken together, the above policy-related findings indicate that creating a business environment that is conducive to recycling magnets would require multiple policy intervals across a range of different EU domains, including the policy portfolio on sustainable products and ecodesign, EU laws on waste shipments and product-specific legislation. The need to employ a variety of policy instruments to support sustainability transitions is shared by various other scholars such as Grubb et al. (2017), Roberts et al. (2023) and Meissner and Kergroach (2021), even if their work does not focus explicitly on the circularity of CRMs.

Concerning the supply chain, the lack of a stable supply of EoL products (e.g. Gregoir et al., 2022), the unpredictability about future waste volumes (e.g. van Nielen et al., 2023), and the uncertainties about permanent magnet use in products (Jönsson et al., 2020) – partly linked to the lack of labels discussed above – were key barriers confirmed by our interview evidence. Our supply chain-related findings however extend beyond these areas by highlighting important challenges stemming from the lack of certain segments of the value chain of RE permanent magnets in the EU and fragmented recycling networks. As also suggested by Mishra et al. (2021) and Rizos et al. (2016), the establishment of business networks and industry alliances can be key levers for distributed collaboration and the discovery of new solutions on the benefit of the full supply chain. Such industry alliances have been developed in the context of the present EU regulatory framework for batteries (Dobrowolski et al., 2021) and solar panels (Chatzipanagi and Jäger-Waldau, 2023).

On the economic front, our empirical results mirror evidence from the literature about the existing challenges caused by the costly extraction of EoL magnets (e.g. Sprecher et al., 2014), volatile RE prices (e.g. Müller et al., 2016), the uncertain profitability of the magnet recycling business (e.g. Raspini et al., 2022) and the competition with cheaper magnets made of virgin materials (e.g. Diehl et al., 2018). At the same time, we provide new insights suggesting that high energy prices, low REE prices, and lack of an established EU market for recycled magnets have a negative impact on the economic feasibility of recycling processes thus creating further uncertainties. In the area of recycling technologies, our study was aligned with the literature in raising the difficult extraction of magnets as a major barrier. This emphasises the role that ecodesign policies can play in facilitating recycling at the EoL stage of products (Dalhammar, 2016; Ortego et al., 2020). We also contribute to the literature by underlining the importance of addressing the gap related to the lack of engineering skills and technical capacity in the EU.

6. Conclusions

Using data collected from industry experts across the full RE permanent magnet value chain as well as academics, our study identifies the major challenges that need to be overcome by the EU to develop a viable recycling chain. The breadth of identified barriers calls for policy action on multiple fronts (EU policy and multilateral) to address current regulatory gaps and create an environment that provides certainty for businesses across different segments of the recycling chain. Action should target both the upstream (e.g. through ecodesign and information requirements) and the downstream (e.g. through targets and financial incentives aimed at recovery of REEs, funding for recycling projects) parts of the value chain. With regard to the latter, increasing financial support for recycling and R&D projects is deemed important

given the uncertain profitability of the magnet recycling business. Such a mix of policy actions would need to be supported by a sound long-term strategy that acknowledges and addresses the existing challenges in a coherent way. At the same time, the strategy should minimise the risks of introducing inconsistent or unrealistic requirements, particularly as the overall policy framework for recycling permanent magnets is still in its infancy.

Several avenues for future research emerge from the study. First, further investigation of existing financial constraints and how the economics of recycling RE magnets in the EU could be improved is recommended. In this context, insights could be drawn from other sectors where such processes have been established (e.g. batteries). Second, combining the perspectives of experts located in the EU and in other parts of the world where the industry has long been established (i.e. China) could be a promising future avenue for similar exploratory research and provide insights about how some of the identified barriers (e.g. lack of multilateral solutions, missing segments in the value chain) could be overcome. Finally, assessing the interplay of different recycling technologies for magnets and how they can co-exist to maximise the potential for recovery of REEs from EoL applications is another suggested area of research.

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CRedit authorship contribution statement

Vasileios Rizos: Writing – original draft, Supervision, Methodology, Investigation, Data curation, Conceptualization, Writing – review & editing. **Edoardo Righetti:** Writing – original draft, Investigation, Data curation, Writing – review & editing. **Amin Kassab:** Writing – original draft, Investigation, Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors do not have permission to share data.

Appendix

Interview guide

I contact you on behalf of the Energy, Resources and Climate Change Unit of CEPIS, a Brussels-based think tank. We are currently conducting research on policy drivers around the recycling of permanent magnets in the context of the INSPIRES EIT RawMaterials project. Our work investigates the barriers (policy, market, technological) to developing a supply chain for recovered/recycled rare earth materials in the EU. Given the work that you doing on the topic we thought it would be fruitful to meet digitally or physically and include your perspectives in our research. The purpose of the interview is to collect empirical data and perspectives on developing a supply chain for recovered rare earth materials in the EU. Empirical data and perspectives for this research will be collected through desk-research and interviews with experts from the value chain (industry and academia). The results of this study may be used for research purposes only. No information within the

reports will be attributed to any specific expert unless the expert gives consent before publication via an email. The interview will be semi-structured and will be based on the questions below:

- I. Please provide a short description of your organisation and activities in the rare earth value chain
- II. What is your position and role in the organisation?
- III. What are, in your view, the key challenges and barriers (policy/financial/technological/market) for developing a value chain for recovered/recycled rare earth elements in the EU?
- IV. Can you identify any areas where policy gaps exist and policy intervention is needed?

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